

Laspeyres and Paasche Price Indices: A Summary

What Are Price Indices?

A **price index** measures how the overall price level of a basket of goods and services changes over time. Price indices are used to:

- Track **inflation** and deflation
- Convert **nominal values to real values**
- Compare **purchasing power** across time periods
- Inform **wage adjustments**, pension benefits, and monetary policy

The two most foundational price index formulas are the **Laspeyres Index** and the **Paasche Index**.

Key Notation

Symbol	Meaning
P^0_i	Price of good i in the base period (period 0)
P^n_i	Price of good i in the current period (period n)
Q^0_i	Quantity of good i in the base period
Q^n_i	Quantity of good i in the current period
Σ	Sum across all goods i

1. Laspeyres Price Index

Definition

The **Laspeyres Index** measures price changes using **base period quantities** as weights. It answers the question:

"How much does it cost today to buy the same basket of goods that was purchased in the base period?"

Formula

$$\text{Laspeyres Index } (L_p) = \left[\frac{\sum (P^n_i \times Q^0_i)}{\sum (P^0_i \times Q^0_i)} \right] \times 100$$

- **Numerator:** Cost of the base-period basket at **current prices**
- **Denominator:** Cost of the base-period basket at **base-period prices**

Step-by-Step Example

An economy produces three goods. Base year = Year 0.

Data:

Good	P ⁰	Q ⁰	P ⁿ	Q ⁿ
Rice	\$2	100	\$3	90
Milk	\$1	200	\$1.50	180
Bread	\$3	50	\$4	60

Step 1 — Calculate Pⁿ_i × Q⁰_i (current prices × base quantities):

Good	P ⁿ × Q ⁰
Rice	\$3 × 100 = \$300
Milk	\$1.50 × 200 = \$300
Bread	\$4 × 50 = \$200
Total	\$800

Step 2 — Calculate P⁰_i × Q⁰_i (base prices × base quantities):

Good	$P^0 \times Q^0$
Rice	$\$2 \times 100 = \200
Milk	$\$1 \times 200 = \200
Bread	$\$3 \times 50 = \150
Total	\$550

Step 3 — Calculate the Index:

$$\begin{aligned}
 L_p &= (\$800 / \$550) \times 100 \\
 &= 1.4545 \times 100 \\
 &\approx 145.45
 \end{aligned}$$

Interpretation: The price level has risen by approximately **45.45%** since the base period, according to the Laspeyres Index.

Bias in the Laspeyres Index

The Laspeyres Index tends to **overstate inflation**. This is because it holds quantities fixed at base-period levels and does not account for the fact that consumers **substitute away** from goods that have become relatively more expensive.

If the price of rice rises sharply, consumers will buy less rice and more of a cheaper substitute — but the Laspeyres Index still assumes they buy the same amount of rice as before, inflating the measured cost.

2. Paasche Price Index

Definition

The **Paasche Index** measures price changes using **current period quantities** as weights. It answers the question:

"How much does it cost today to buy the current basket of goods, compared to what that same basket would have cost in the base period?"

Formula

$$\text{Paasche Index } (P_p) = \left[\frac{\sum (P^n_i \times Q^n_i)}{\sum (P^0_i \times Q^n_i)} \right] \times 100$$

- **Numerator:** Cost of the **current basket** at **current prices**
- **Denominator:** Cost of the **current basket** at **base-period prices**

Step-by-Step Example

Using the same data as above:

Step 1 — Calculate $P^n_i \times Q^n_i$ (current prices $\times$ current quantities):

Good	$P^n \times Q^n$
Rice	$\$3 \times 90 = \270
Milk	$\$1.50 \times 180 = \270
Bread	$\$4 \times 60 = \240
Total	\$780

Step 2 — Calculate $P^0_i \times Q^n_i$ (base prices $\times$ current quantities):

Good	$P^0 \times Q^n$
Rice	$\$2 \times 90 = \180
Milk	$\$1 \times 180 = \180
Bread	$\$3 \times 60 = \180
Total	\$540

Step 3 — Calculate the Index:

$$\begin{aligned} P_p &= (\$780 / \$540) \times 100 \\ &= 1.4444 \times 100 \\ &\approx 144.44 \end{aligned}$$

Interpretation: The price level has risen by approximately **44.44%** since the base period, according to the Paasche Index.

Bias in the Paasche Index

The Paasche Index tends to **understate inflation**. Because it uses current quantities — which already reflect consumer substitution toward cheaper goods — it underweights goods whose prices rose the most.

If consumers switched from expensive rice to cheaper bread, the Paasche Index gives more weight to bread (which is now bought more), masking how much the overall price level actually rose.

3. Comparing Laspeyres and Paasche

Results from the Same Example

Index	Value	Inflation Estimate
Laspeyres	145.45	45.45%
Paasche	144.44	44.44%

As expected, **Laspeyres > Paasche** — this is the typical relationship and is known as the **substitution effect bias**.

Head-to-Head Comparison

Feature	Laspeyres	Paasche
Quantity weights	Base period (Q^0)	Current period (Q^n)
Data requirement	Only base-period quantities needed	Requires updated quantities each period
Bias direction	Overstates inflation	Understates inflation
Ease of calculation	Easier — weights fixed over time	Harder — weights change every period
Common use	CPI (Consumer Price Index)	GDP Deflator

Feature	Laspeyres	Paasche
Typical result	Higher index value	Lower index value

4. The Fisher Ideal Index

Because Laspeyres overstates and Paasche understates inflation, economists often use the **Fisher Ideal Index** as a compromise — it is the **geometric mean** of both.

Formula

$$\text{Fisher Index (F}_p\text{)} = \sqrt{\text{Laspeyres} \times \text{Paasche}}$$

Example

$$\begin{aligned} F_p &= \sqrt{145.45 \times 144.44} \\ &= \sqrt{21,011.81} \\ &\approx 144.95 \end{aligned}$$

Interpretation: Prices rose by approximately **44.95%** — splitting the difference between the two indices.

The Fisher Index is considered the most **theoretically accurate** price index because it accounts for substitution in both directions.

5. Real-World Applications

Index	Real-World Example
Laspeyres	Consumer Price Index (CPI) in most countries
Paasche	GDP Deflator used in national accounts
Fisher	Used in U.S. National Income and Product Accounts (NIPA)

6. Using Price Indices to Deflate Nominal Values

Once a price index is calculated, it can be used to convert nominal values to real values:

$$\text{Real Value} = (\text{Nominal Value} / \text{Price Index}) \times 100$$

Example: If nominal spending is \$900 and the Laspeyres Index is 145.45:

$$\begin{aligned} \text{Real Value} &= (\$900 / 145.45) \times 100 \\ &\approx \$618.80 \end{aligned}$$

This means \$900 today has the same purchasing power as approximately \$618.80 in the base year.

Quick Reference

Metric	Formula
Laspeyres Index	$[\Sigma(P^n \times Q^0) \div \Sigma(P^0 \times Q^0)] \times 100$
Paasche Index	$[\Sigma(P^n \times Q^n) \div \Sigma(P^0 \times Q^n)] \times 100$
Fisher Ideal Index	$\sqrt{(\text{Laspeyres} \times \text{Paasche})}$
Deflating Nominal Values	$(\text{Nominal Value} \div \text{Price Index}) \times 100$
Inflation Rate from Index	$((\text{Index_current} - \text{Index_base}) \div \text{Index_base}) \times 100$